



Mobile Forensics



Definition: Mobile Forensics is the branch of digital forensics that focuses on **recovering and analyzing data from smartphones, tablets, and SIM cards.**

It helps investigators extract evidence like **call logs, WhatsApp chats, SMS, GPS location, images, browsing history, app data, and emails.**

◆ What is Extracted in Mobile Forensics?

1. **Call Logs** → Incoming, outgoing, missed calls, duration.
2. **SMS/MMS** → Deleted and existing text messages.
3. **WhatsApp / Other Apps** → Chats, media, documents, voice notes.
4. **GPS Data** → Location history, navigation routes, geotagged photos.
5. **Contacts & Emails** → Address book, Gmail/Outlook sync data.
6. **Multimedia** → Photos, videos, voice recordings.
7. **System Files** → Device info, IMEI, installed apps.

◆ How Mobile Forensics is Done (Process)

1. Seizure & Preservation

- Secure the device (put it in **Faraday Bag** to block signals).
- Prevent remote wiping or tampering.

2. Acquisition (Data Extraction)

- **Logical Extraction** → Extracts files that are visible (contacts, SMS, call logs).
- **Physical Extraction** → Bit-by-bit copy of the phone's memory, including deleted data.
- **File System Extraction** → Extracts file system structure (app data, images, logs).
- **Cloud Extraction** → Extract backups from iCloud, Google Drive, WhatsApp Cloud.

3. Analysis

- Recover deleted chats, SMS, media.
- Correlate GPS data with timestamps.
- Timeline of calls, messages, activities.

4. Reporting

- Prepare legally valid report with findings.

◆ Tools Used in Mobile Forensics

- Free / Open-source

- **Autopsy (with plugins)** → Basic analysis.
- **Andriller** → Extracts Android data.
- **MobSF (Mobile Security Framework)** → App analysis.